

$$\Delta U = 3.55 - 3.0$$

- 16 A** Constant pressure compression – in the p - T diagram it is still a horizontal line. $pV = nRT$. Since temperature decreases in a constant pressure compression, it is represented by a horizontal line from right to left.

Constant temperature expansion – Temperature is constant and pressure decreases. It is represented by a downward vertical line in the p - T diagram.

Constant volume process – From $pV = nRT$, since V is constant, the graph of p against T is a straight line passing through the origin. This is why option D is wrong.

17 D Since temperature is constant $\Delta U = 0$